Bidirectional Guided Attention Network for 3-D Semantic Detection of Remote Sensing Images

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***Abstract*— Semantic segmentation and disparity estimation are in the research frontier of the computer vision and remote sensing (RS) fields. However, existing methods mostly deal with these two problems separately or use a combination of multiple models to solve these two tasks. Due to a lack of sufficient information sharing and fusion, they still have difficulties in coping with seasonal appearance differences in 3-D RS problems. In this article, we propose a novel multitask learning architecture that considers the bottom–up and up–bottom visual attention mechanism for 3-D semantic detection, named bidirectional guided attention network (BGA-Net). BGA-Net consists of five modules: unified backbone module (UBM), bidirectional guided attention module (BGAM), semantic segmentation module (SSM), feature matching module (FMM), and bidirectional fusion module (BFM). First, in UBM, we use a shared backbone to extract unified features and share them with three branches/modules (BGAM, SSM, and FMM). Then, SSM and FMM branches are applied to estimate segmentation and disparity maps, whereas the third branch/module (BGAM) shares the global features to guide the task-specific learning via attention mechanism. Finally, we fuse the results of the two tasks by BFM to improve the final performance. Extensive experiments demonstrate that: 1) our BGA-Net can handle the two tasks simultaneously and can be trained in an end-to-end way; 2) these modules fully take advantage of the two tasks' information to share features and enhance the scene understanding ability, effectively against seasons change of RS images; and 3) BGA-Net has notable superiority and greater flexibility and also sets a new state of the art on the urban semantic 3-D (US3D) benchmark. Moreover, BGA-Net also provides insights into the intelligent interpretation of RS data images.**

***Index Terms*— Bidirectional guided aggregation, remote sensing image, semantic segmentation, stereo matching, visual attention mechanism.**

I. Introduction

SEMANTIC segmentation and stereo matching are fundamental problems in computer vision and remote sensing (RS) fields, which are essential in earth observation [1]–[3], scene understanding [4], [5], autonomous driving [6], [7], 3-D model reconstruction [8]–[10], and object recognition [11]–[13]. To adapt more complex real-world applications, particularly in 3-D model reconstruction of RS images, multitask learning (MTL) [14] are recently used to perform multiple tasks. In this article, a novel MTL architecture with the bottom–up and up–bottom visual attention mechanism for 3-D semantic detection, named bidirectional guided attention network (BAG-Net), is proposed for 3-D semantic detection of RS images, which can generate the segmentation and disparity maps simultaneously.

Semantic segmentation and stereo matching as classical vision single-task learning problems have been studied for many years [1], [15]. With the wave of deep learning in artificial intelligence (AI), many researchers have designed straightforward and advanced networks to improve the performance and made excellent progress for the two tasks [5], [16]. To better understand the research background, the single-task learning for segmentation or stereo matching will be briefly overviewed in the following paragraphs.

For the semantic segmentation task, many methods were proposed by viewing it as a classification problem, which is to assign the category label for each pixel in the image [17], [18]. To achieve accurate scene perception, the prior information of scene context or the filters with larger receptive fields is employed to build the knowledge graph [4], [18]. On this basis, the proposed methods are made use of various visual attention mechanisms (e.g., channel attention, self-attention, and so on) for helping the network to detect the object in an image pattern [19], [20]. These approaches boost dynamic object detection and scene understanding.

For the stereo matching task, most methods were addressed by viewing it as a regression problem, aiming to achieve pixelwise dense prediction from the stereo images [15], [21]. Compared with the traditional approaches, they used the deep features extracted from convolutional neural networks (CNNs) to exploit inherent global information, leading to the performance of disparity estimation improved significantly [22], [23]. On this basis, this task was formulated as supervised learning based on the labeled data sets [23] or treated as unsupervised learning via constructing photometric loss for disparity prediction [22]. These methods for both the semantic segmentation task and the stereo matching task have promoted the development of 3-D scene understanding and produced many state-of-the-art results.

In this article, we try to integrate the two tasks into one, called 3-D semantic detection, with emerging MTL. Most current approaches for the two tasks focus on how to exploit

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semantic or disparity information to improve the accuracy of either task, instead of predicting the disparity and segmentation map at the same time [2], [3], [24]. Moreover, existing stereo methods work on the standard stereo images, which are collected simultaneously. However, the incidental satellite collects the "left" and "right" images in the near position but at different times, which causes seasonal appearance differences to pose a challenge for stereo matching, as shown in Fig. 1.

To solve the abovementioned problems by using the MTL, we proposed a BAG-Net for 3-D semantic detection, which uses the bottom–up and up–bottom visual attention mechanism. First, we employ a single shared module to learn unified features across all tasks. Then, for each task, we adopt the bidirectional attention mechanism to construct the bridge for strengthening the links between the two tasks. Moreover, this mechanism allows the learning of both task-shared and task-specific features in an end-to-end manner. At the end of our model, we apply two fusion modules to fuse the disparity and segmentation maps for further improving the accuracy of each task. The experimental results indicate that our method has better robustness to each task and can handle seasonal appearance differences well via the fusion and sharing of task-specific features. Our major contributions in this article are as follows.

1) We propose an end-to-end MTL framework (the bidirectional guided attention network (BGA-Net) for 3-D semantic detection without any post-processing.
2) We develop a global context extraction strategy for constructing the bidirectional attention mechanism, which enhances the connection between each task to improve two tasks' performance.
3) Our model can achieve state-of-the-art results for the two tasks simultaneously. Furthermore, it is more robust to seasonal appearance differences and reaches top performance on the urban semantic 3-D (US3D) data set.

II. Related Work

In the past, computer vision methods based on deep learning often regard the stereo matching and semantic segmentation as two tasks separately. Recently, the rapid progress in the two tasks made it possible to achieve disparity estimation and segmentation at a finer level. Meanwhile, the development of MTL boosted the performance of 3-D scene understanding. In this work, we aim at bridging the relationship between stereo matching and semantic segmentation. According to our work, we briefly review the existing methods as follows.

A. Semantic Segmentation

The past several years have witnessed great progress on semantic segmentation after a fully convolutional network (FCN) [17], [18]. Many methods exploit the contextual information by multiscale information and the encoder–decoder model [25], [26], and the latest approaches employ a new technique called attention to realize high-level segmentation [17], [27].

Long *et al.* [25] proposed a groundbreaking method called the FCN, which rapidly promoted the development of semantic segmentation. He *et al.* [13] proposed a multiscale 3-D CNN for hyperspectral image classification. Ronneberger *et al.* [27] utilized the CNN to build an

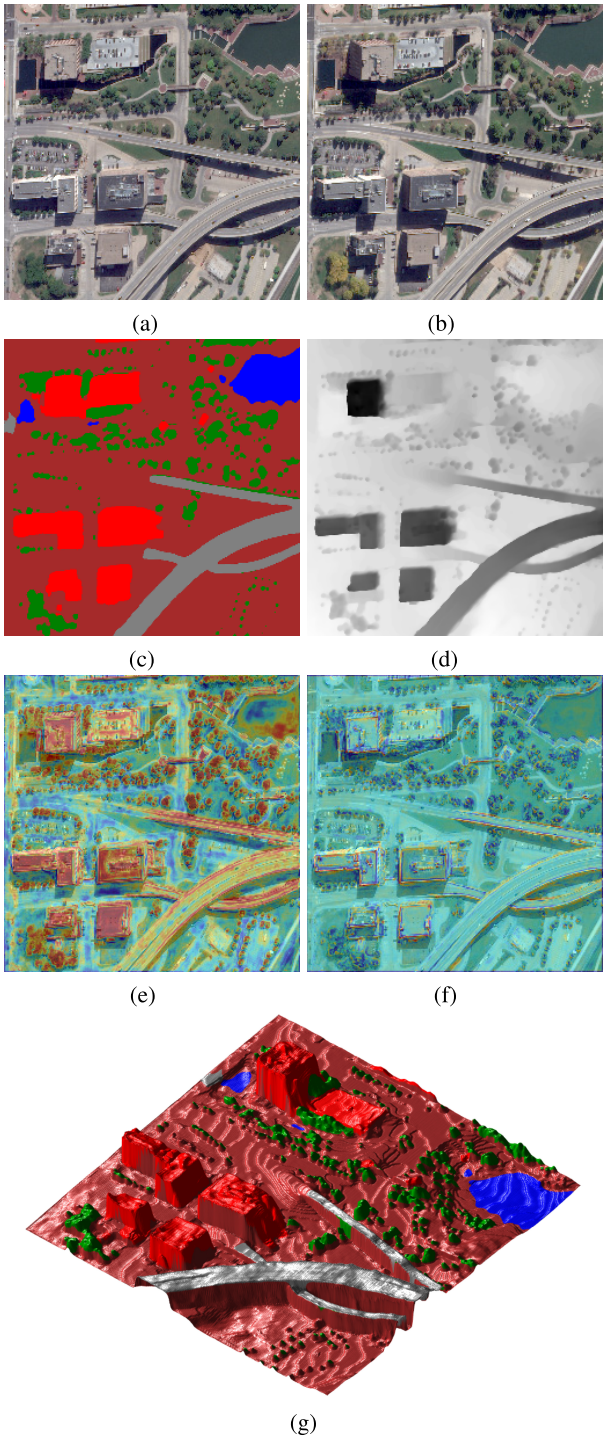

Fig. 1. Results on the US3D data set. Given (a) and (b) stereo image pairs, the goal of 3-D semantic detection is to predict (c) segmentation map/CLS and (d) disparity map/DSP at the pixel level. We present the CLS and DSP attention maps generated from our network. Note that (e) our CLS attention map can grasp the target object in the image precisely and (f) DSP attention map could predict the potential object for a disparity value. Furthermore, we reconstruct the 3-D scene as shown in (g). It demonstrates that our model can be applied in (g) large 3-D scene reconstruction of RS images. Here, CLS denotes the classification results of semantic segmentation and DSP means disparity results of stereo matching.

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encoder–decoder architecture, which could be trained in an end-to-end way from a few images and outperformed the previous methods. Chen *et al.* [18], [26] developed the DeepLab series around how to apply atrous spatial pyramid pooling (ASPP) for collecting contextual information, which achieved the top performance in an end-to-end framework. Lin *et al.* [28] exploited the potential of the inherent multiscale and pyramidal hierarchy to construct feature pyramids, demonstrating that it was still critical to explicitly address multiscale problems using pyramid representations. On this basis, Kirillov *et al.* [17] designed a single network for panoptic segmentation yielding top-performing.

B. Stereo Matching

With the development of deep learning, many researchers take advantage of the deep CNNs to improve accuracy and running time. The central idea is that how to use CNNs to compare the patchwise similarity for obtaining the matching score and then get the final disparity maps via regression or classification process [16], [21], [24].

Kendall *et al.* [15] introduced a 3-D CNN to learn geometry and context information obtaining good results on several benchmarks. Chang and Chen [16] used the spatial pyramid pooling (SPP) and a stacked hourglass module to further improve accuracy. Zhong *et al.* [22] focused on stereo video and applied traditional matching algorithms as loss function to achieve unsupervised learning. Zhang *et. al* [23] used a semiglobal aggregation layer and a local guided aggregation layer to replace the widely used 3-D convolutional layer for reducing the memory-consuming and computational cost. Tonioni *et al.* [29] introduced a new lightweight architecture based on an online-learning strategy for addressing the problem of computational demanding and unreal time.

C. Attention-Based Modules

Attention-based modules have been widely applied in visual tasks, which aim at driving the network paying close attention to the region of interest (ROI) [30]. Recently, many approaches have promoted the development of the attention-and-gating mechanism, and they adopted this mechanism to formulate channelwise relationships [19], [20], [31].

Li *et al.* [19] introduced the attention-based module to embed in feature pyramid networks (FPNs) [28] for improving the background and foreground understanding ability. Fu *et al.* [32] adopted the self-attention mechanism to model the semantic interdependencies in spatial and channel dimensions for further improving feature representation. Wang *et al.* [31] bridged self-attention for machine translation to achieve long-range response at a position in the video classification task. Cao *et al.* [20] simplified the nonlocal block (NL-block) and combined it with squeeze-and-excitation networks (SE-Nets) [30] to construct the global context block (GC-block), which proved that it can effectively extract the global context with significantly fewer computation.

D. Multitask Learning

MTL is a recently proposed method that aims to do two or more tasks in a model. Compared with standard single-task learning, the key to training multiple tasks is how to share the feature representation and balance different loss functions in the multiple tasks [14].

Liu *et al.* [14] applied the task-shared or task-specific way to learn a generalizable representation and adopted the loss weights to enable learning of all tasks with equal importance. Yang *et al.* [24] conducted the semantic information embedding into the network to improve the performance in the occlusion region. Qin *et al.* [2] viewed MTL as a standard single-task learning but using the fusion way to improve the performance for each task. Chen *et al.* [3] enabled much more expressive combinations of features to improve the learning ability for generalization across tasks.

In this work, we proposed a novel network with a three-branch structure and the bidirectional visual attention mechanism for 3-D semantic detection, named the BGA-Net. In our model, two of the three branches are used to predict the results of different tasks, and the last branch applies the attention mechanism as a bridge to the two branches for enhancing the connection between them. The results show that our model can be trained in an end-to-end manner and estimated as the state-of-the-art results for the two tasks simultaneously.

III. Our Method

In this section, we describe our proposed BGA-Net, and the whole architecture is clearly shown in Fig. 2 with the Appendix. Our BGA-Net consists of a shared backbone module, three branches, and two fusion modules. First, we use a unified backbone module (UBM) to extract the unified feature map, and this way is helpful to obtain a more general feature map, as shown in Section III-A. Next, we employ semantic segmentation and feature matching modules (FMMs) to do specific tasks and apply a bidirectional attention module to share the global feature for guiding task-specific learning, as described in Sections III-B–III-D. Then, we adopt a bidirectional fusion module (BFM) to fuse the segmentation map and disparity map for further improving the performance, as discussed in Section III-E. Finally, we introduce our loss functions for each task, as presented in Section III-F. For the convenience of description, we assume that the input resolution of the stereo pairs is $H \times W$, and the basic number F of convolution filter is 32.

A. Unified Backbone Module

To bridge the gap between two tasks, we apply the UBM to extract features from the left–right image pair, as shown in Fig. 2. Compared with our previous approach [33], [34], the proposed UBM fuses two models (the feature extraction modules of semantic segmentation and stereo matching) together by sharing the same backbone and generates the left–right features from parallel branches.

The UBM consists of two parts, a unary feature extraction part and a multiscale features fusion part, as shown in Fig. 3. The structure of UBM is very similar to the 2-D feature extraction in our previous stereo matching approach, except for

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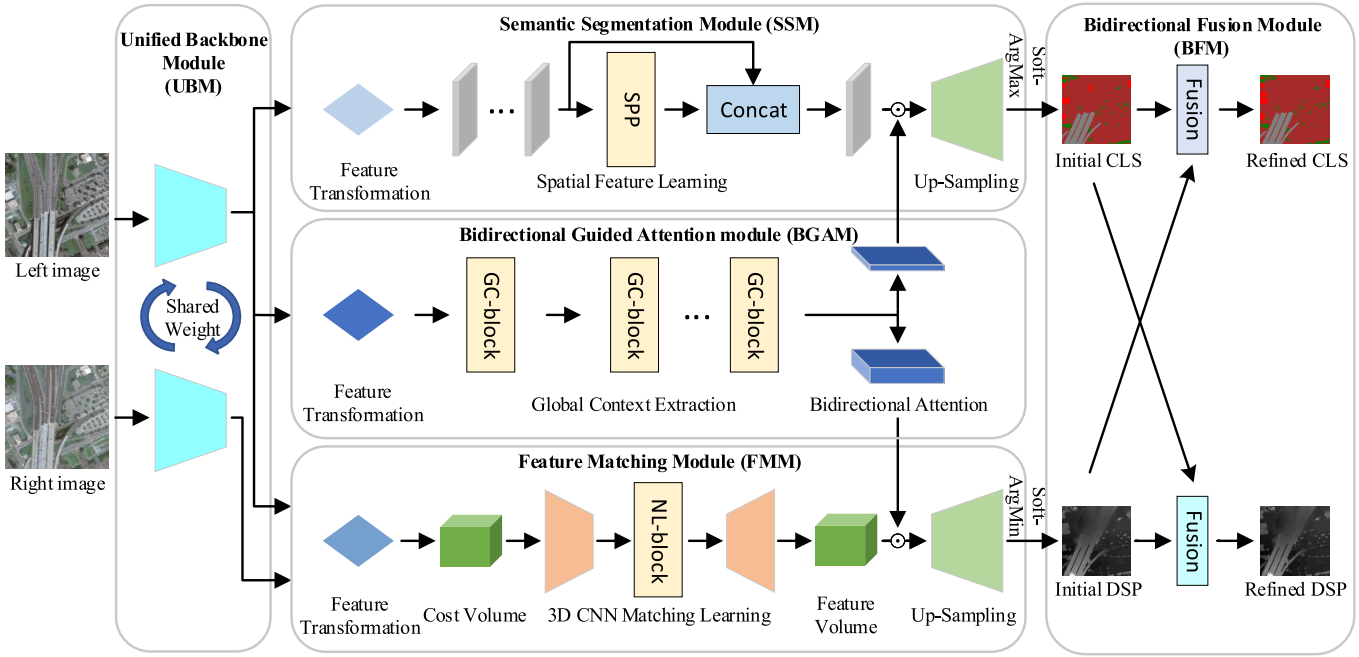

Fig. 2. Our BGA-Net. The network consists of five modules: unified backbone module, bidirectional attention module, SSM, FMM, and BFM.

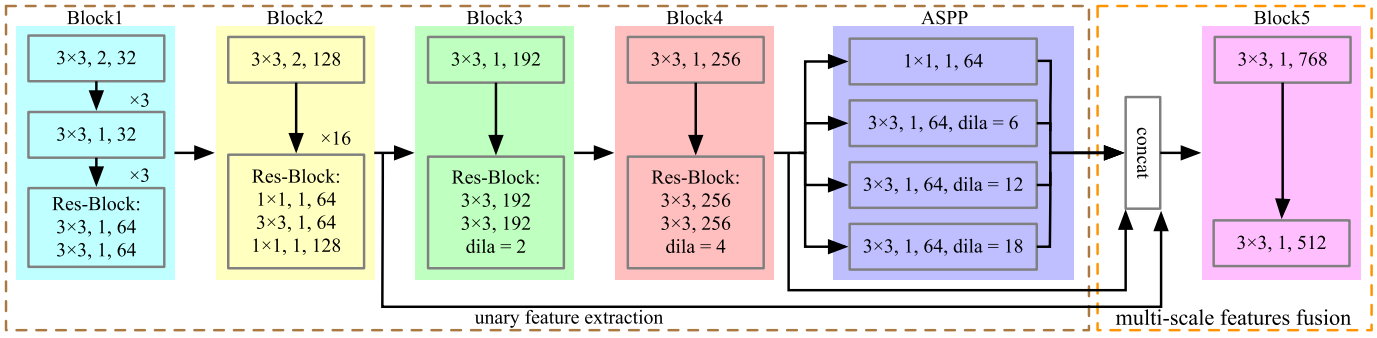

Fig. 3. Structure of UBM. The module consists of the unary feature extraction part and the multiscale features fusion part.

adding the dimension of feature map and replacing SPP [35] with ASPP [26].

1) For adding the dimension, we consider that the multi-tasks need more quantity of feature maps to meet the demand of multibranches.
2) For using ASPP instead of SPP in this step, we hope that the UBM could obtain a larger receptive field by network learning, instead of mechanically shrinking space via pooling operation.

After UBM, we could get the left–right unified feature maps $X_{l,r} \in \mathbb{R}^{H/4\times W/4\times 16F}$ from the left–right images, respectively.

B. Bidirectional Guided Attention Module

Considering the complementary relationship between the segmentation map and disparity map, we introduce a bidirectional guided attention module (BGAM) as a bridge between them. Different from traditional attention approaches, which only focus on one task, the proposed BGAM extracts the global context information and facilitates the performance of two tasks. The BGAM consists of three parts: the feature transformation part, the global context extraction part, and the bidirectional attention part.

1) Feature Transformation Part: The first step of BGAM is to reduce the feature dimensions by a series of 2-D convolutional operations. We consider that the BGAM aims to obtain global information from the unified feature maps X_l to generate the attention maps. Therefore, we first adopt a simple strategy to convert the unified feature maps via reducing the feature dimension. The BGAM applies one 256-channel and one 128-channel 2-D convolutional layers with 1×1 filters to reduce the feature dimension and then utilizes one 128-channel residual blocks with 3×3 filters to regularize the feature maps. After this step, we could get the initial global information maps with the size $1/4\ H \times 1/4\ W \times 4F$.

2) Global Context Extraction Part: The second step of BGAM is to capture global context information by a group of GC-blocks. Our previous works [34] and GC-Net [20] both verify that NL-block [31] can model the semantic interdependencies in the different application scenarios.

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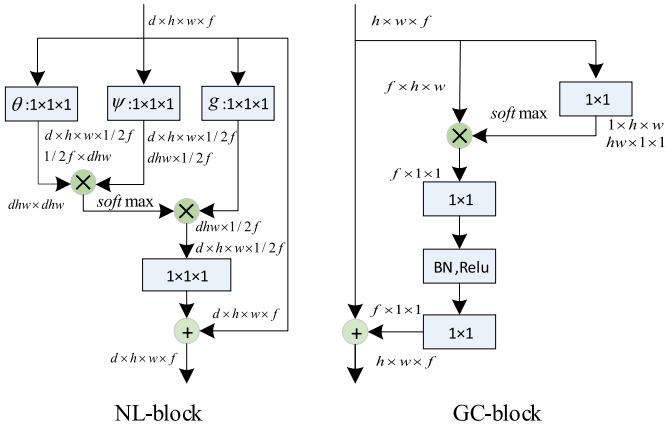

Fig. 4. Architecture of GC-block and NL-block. The feature maps are shown as feature dimensions, e.g., $d \times h \times w \times f$ represents feature maps with disparity d, height h, width w, and channel number f. Here, $\otimes$ means the matrix multiplication and $\oplus$ denotes the elementwise sum.

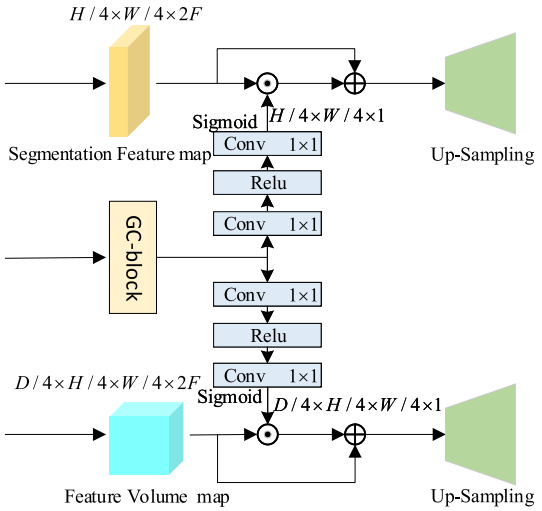

Fig. 5. BGAM is designed to establish the complementary relationship. We adopt this block to generate the attention maps for guiding the SSM and FMM. Here, $\oplus$ denotes the elementwise sum and $\odot$ denotes the spatial elementwise multiplication.

GC-Net [20] creates a simplified NL-block called GC-block, and the GC-block has been demonstrated better effect to capture the global contextual information with significantly less computation in object detection. The NL-block and GC-block are shown in Fig. 4; we follow those works and adopt a group of GC-block to capture global context information from the initial global information maps. After this step, we could obtain the global context maps $G \in \mathbb{R}^{1/4\ H\times 1/4\ W\times 4F}$.

3) Bidirectional Attention Part : The final step of BGAM is to produce the bidirectional attention maps for semantic segmentation and stereo matching via the sigmoid function and 2-D convolutional operations. The global context maps contain the complete query-independent context information [20]. Those maps have the potential of converting itself to the semantic attention map or disparity attention maps via 2-D convolutional operations, and we will prove it in the experiments. Next, we explain the relationship between the semantic attention maps and the disparity attention maps.

The attention operation aims to observe the target object. For the semantic segmentation, the semantic attention map will assign more weights to the objects (e.g., building, road, and tree) and fewer weights to the background. For the stereo matching, each object is viewed as a rigid surface in general; thus, the same object has the same or close disparity d. The disparity attention maps will assign different weights for each disparity d from 0 to $D/4$, where D means the max disparity value. This process could be seen as allocating each object to the different disparity d.

From the above description, the semantic and disparity attention maps could be viewed as two special types from global context maps. Therefore, we can use the convolutions to encode global context maps to the semantic and disparity attention maps, and this strategy can link the semantic segmentation module (SSM) and FMM. As shown in Fig. 5, we utilize the global context maps from a group of GC-block for bidirectional attention operation.

To further understand the bidirectional attention mechanism, we give a detailed formulation for this process. Given the global context maps $G \in \mathbb{R}^{H/4\times W/4\times 4F}$ from the GC-blocks, the coarse semantic segmentation attention map M before the sigmoid activation can be formulated as

$$M = \mathcal{F}(\sigma(\mathcal{F}(G, w_1)), w_2) \tag{1}$$

where $\mathcal{F}$ denotes a convolution function, σ represents the ReLU function, $M \in \mathbb{R}^{H/4\times W/4\times 1}$ means the generated coarse semantic segmentation attention map, and $w_1 \in \mathbb{R}^{4F\times 1\times 1\times 2F}$ and $w_2 \in \mathbb{R}^{2F\times 1\times 1\times 1}$ indicate convolutional parameters.

To convert the weight of M into the range from 0 to 1, we formulate the semantic segmentation attention map M' as sigmoid(M). Then, the activated feature map S' in SSM can be presented as

$$S' = S \odot M' \oplus S \tag{2}$$

where $\odot$ and $\oplus$ denote the elementwise multiplication and sum, respectively, and S denotes the semantic feature map produced by SSM.

In the same way, we follow (1) and adopt $w_3 \in \mathbb{R}^{2F\times 1\times 1\times D/4}$ to replace w_2 for getting the coarse disparity attention maps $R \in \mathbb{R}^{H/4\times W/4\times D/4}$. Then, we assign R for each disparity d from 0 to $D/4$, which follows (2) but in the four dimensions (disparity $\times$ height $\times$ width $\times$ feature). Thus, we obtain the activated feature map in the FMM.

Based on the above formulation, we highlight the object in semantic segmentation and FMMs via attention operation. It also promotes the learning of the foreground and background in turn by enhancing the weights of activated foreground regions during backpropagation.

C. Semantic Segmentation Module

In this module, we apply a series of 2-D convolutional operations to generate the initial segmentation map. The multiscale information has been proved to promote scene understanding for semantic segmentation. Thus, we use the famous SPP block to extract multiscale information. The SSM contains three

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parts: feature transformation part, spatial feature learning part, and upsampling part.

1) Feature Transformation Part: The first step of SSM is to reduce the feature dimensions by a series of 2-D convolutional operations. To achieve this goal, we adopt the same strategy that is similar to the same part in BGAM. After this step, we could get the initial semantic feature maps with the size $H/4 \times W/4 \times 4F$.

2) Spatial Feature Learning Part: The second step of SSM is to extract the multiscale features from the initial semantic feature maps. To further extract hierarchical contextual information, we first apply sixteen 128-channel bottleneck blocks to get the deep semantic feature maps and then adopt an SPP to separates them into different subregions. Next, we concatenate hierarchical contextual information and deep semantic feature maps and then fuse them via one 128-channel and one 64-channel 2-D convolutional operations with 3×3 filters to obtain the semantic feature map S. After that, the semantic feature map S is guided by the semantic segmentation attention map M' to get the activated feature map S', as shown in Section III-B. After this part, we obtain the activated feature map $S' \in \mathbb{R}^{H/4 \times W/4 \times 2F}$.

3) Upsampling Part: The final step of SSM is to recover the size of the activated feature map S' for producing the high-quality score map. We first utilize one 32-channel and one C-channel 2-D deconvolutional layers (both the stride is 2) to recover the size for getting the score map $S_{\text{init}} \in \mathbb{R}^{H \times W \times C}$, where C represents the number of categories. Then, we apply one C-channel 2-D convolutional layer to regularize the score map for improving the quality. With the ArgMax process, we obtain the initial segmentation map $\hat{S}_{\text{init}} \in \mathbb{R}^{H \times W}$.

D. Feature Matching Module

In this module, we apply a series of 2-D and 3-D convolutional operations to produce the initial disparity map. Disparity estimation aims to predict the accurate disparity map from the left–right images $I_{\text{l,r}}$. Therefore, we first pass the left–right images to UBM for getting the left–right unified feature maps $X_{\text{l,r}}$ of the left–right images $I_{\text{l,r}}$. Then, we feed the left–right unified feature maps $X_{\text{l,r}}$ into the FMM for generating the initial disparity map. FMM contains four parts: feature transformation part, cost volume construction part, stereo matching part, and disparity regression part.

1) Feature Transformation Part: The first step of FMM is to reduce the feature dimensions by a series of 2-D convolutional operations. Different from the same part in BGAM or SSM, the matching process needs a lot of memory. Therefore, we should reduce more dimensions than the module mentioned earlier. The FMM adopts one 128-channel and one 64-channel 2-D convolutional layers with 1×1 filters to reduce the feature dimension and then utilizes one 64-channel residual block with 3×3 filters to regularize the feature maps. In this way, we obtain the left–right transformed feature maps $f_l, f_r \in \mathbb{R}^{H/4 \times W/4 \times 2F}$.

2) Cost Volume Construction Part: The second step of FMM is to aggregate left–right transformed feature maps f_l, f_r by the variance-based cost metric $\mathcal{M}$. We consider this cost metric as the mapping $\mathcal{M}: \underbrace{\{f_l, f_{\text{r},1}\}, \ldots, \{f_l, f_{r,D/4}\}}_{D/4} \to \mathcal{V}$ that

$$\begin{aligned}\mathcal{V} &= \mathcal{M}\big(\{f_l, f_{\text{r},1}\}, \ldots, \{f_l, f_{r,D/4}\}\big) \\ &= \text{stack}\left(\frac{\left(f_l{}^2 - \overline{f_i}\right)^2 + \left(f_{r,i} - \overline{f_i}\right)^2}{2}\right) \end{aligned} \tag{3}$$

where $f_{r,i}$ denotes the right transformed feature map f_r traversed with a preset disparity i, $\overline{f_i}$ means the average volume of f_l and $f_{r,i}$, and all operations above are elementwise. After this part, we get cost volume $\mathcal{V} \in \mathbb{R}^{D/4 \times H/4 \times W/4 \times 2F}$.

3) Stereo Matching Part: The third step of FMM is to regularize the cost volume by a series of 3-D convolutional operations. In our previous work [34], NL-block as a self-attention method improves performance significantly in matching tasks. Therefore, we continue to employ this design except for the number of NL-block and add the attention mechanism to guide the matching cost aggregation, as shown in Section III-B. Because this structure is not critical in this article, we suggest that interested scholars read the previous paper [34] or the Appendix to get detailed information. Thus, we obtain the regularized volume $\mathcal{V}' \in \mathbb{R}^{D \times H \times W}$.

4) Disparity Regression Part: The final step of SMM is to regress the initial disparity map $\hat{d}_{\text{init}}$ by the ArgMin operation. First, the probability of each disparity d is calculated from the regularized cost volume $\mathcal{V}'$ via the softmax operation. Next, the predicted initial disparity $\hat{d}_{\text{init}}$ is calculated as the sum of each disparity d weighted by its probability. The regression process can be expressed as

$$\hat{d}_{\text{init}} = \sum_{d=0}^{D} d \times \text{softmax}\big(-\mathcal{V}'\big). \tag{4}$$

After that, we obtain the initial disparity map $\hat{d}_{\text{init}} \in \mathbb{R}^{H \times W}$.

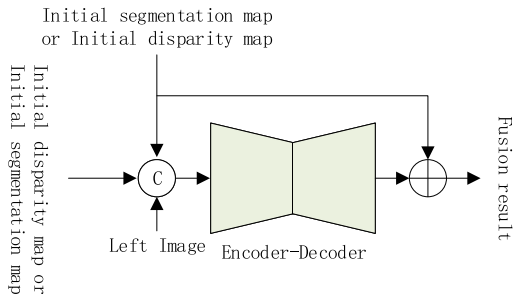

Fig. 6. Fusion module. The module contains two parts: semantic fusion and disparity fusion. Note that semantic fusion and disparity fusion are only different in the last layer.

E. Bidirectional Fusion Module

In this module, we adopt a series of 2-D convolutional operations to fuse the segmentation and disparity maps to improve accuracy further. BFM contains two parts, the semantic fusion part and the disparity fusion part, as shown in Fig. 6.

1) Semantic Fusion Part: In the semantic fusion part, we apply a residual learning network at the end of BGA-Net. We first concatenate the initial score map S_{init}, the left

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input image, and the initial disparity map as a $(4 + C)$-channel input. To learn the segmentation residual, we apply the downsampling process to encode the input data and the upsampling process to decode the encoded feature maps. After that, we employ one C-channel 2-D convolutional operation to get the segmentation residual. The refined score map is calculated by adding the residual and the initial score map together, and then, the refined segmentation map is obtained by the ArgMax operation.

2) Disparity Fusion Part: In the disparity fusion part, the structure is similar to the semantic fusion part except for the last layer. The last layer in this section uses one 1-channel convolutional operation to replace one C-channel 2-D convolutional operation, largely reducing the computing.

After the fusion module, we obtain the refined score map and disparity map with the same size as the input images.

F. Loss Function

In the previous sections, we regard the stereo matching as a regression problem and view the semantic segmentation as a classification problem. Here, we detail the loss functions in these two conditions. Then, we fuse the two types of loss functions for end-to-end training.

1) Loss Function in Semantic Segmentation: Since semantic segmentation is a classical classification problem, we use the cross-entropy function to learn the scene understanding. Given estimated score map, we convert it to the probability volume $\mathcal{P}$ via the softmax operation. Then, we adopt the cross-entropy function to regularize the probability volume $\mathcal{P}$. The process could be represented as

$$\mathcal{L}_{\text{seg}} = \sum_{\mathcal{P}} \left(\sum_{i=1}^{C} -\mathcal{P}(i, p) \cdot \log Q(i, p) \right) \tag{5}$$

where p denotes the spatial image coordinate, i represents the category number, $\mathcal{P}(i, p)$ means a voxel in the probability volume $\mathcal{P}$, Q indicates the ground-truth binary occupancy volume, which is generated by the one-hot encoding of the ground truth with the number of categories C, and $Q(i, p)$ denotes the corresponding voxel to $\mathcal{P}(i, p)$.

2) Loss Function in Stereo Matching: Because stereo matching is a regression problem in this article, we utilize the L_1 loss function to learn the disparity prediction. Because the ground truth of disparity map is sparse set in general, we only average our loss over the labeled pixels. The process could be represented as

$$\mathcal{L}_{\text{dsp}} = \frac{1}{N} \sum_{n=1}^{N} ||d(p) - \hat{d}(p)||_1 \tag{6}$$

where N denotes the number of the labeled pixels, p represents the spatial image coordinate, $hatd(p)$ indicates the value of predict disparity at the pixel p, and $d(p)$ means the value of ground truth at the pixel p.

As mentioned earlier, we employ the $\mathcal{L}_{\text{seg}}$ loss function to calculate the initial and refined score maps for segmentation learning and adopt the $\mathcal{L}_{\text{dsp}}$ loss function to estimate the initial and refined disparity maps for matching learning. Therefore, we can obtain the segmentation loss $\mathcal{L}_1, \mathcal{L}_2$ for the initial and refined score maps and the disparity loss $\mathcal{L}_3, \mathcal{L}_4$ for the initial and refined disparity maps. To balance learning of different loss branches, we introduce loss weights λ_1 and λ_4 for them. Thus, the total loss $\mathcal{L}$ is expressed as

$$\mathcal{L} = \lambda_1 \mathcal{L}_1 + \lambda_2 \mathcal{L}_2 + \lambda_3 \mathcal{L}_3 + \lambda_4 \mathcal{L}_4. \tag{7}$$

IV. Experiments

In this section, we will evaluate the performance of our method on the semantic stereo data sets: Cityscape [36], KITTI [7], and the US3D [37]. First, we present our implementation details about the training and testing sets, as shown in Section IV-A. Next, we verify the contribution of the different components in BGA-Net on the cityscape data set, as shown in Section IV-B. Then, we quantize our method's performance and compare it with state-of-the-art methods on the KITTI and US3D data sets, as shown in Sections IV-C and IV-D. Finally, we analyze our attention mechanism of two tasks and season challenge of stereo matching to prove the effectiveness of our approach, as shown in Sections IV-E and IV-F. Moreover, we will release the code and configurations at Github [1].

A. Implementation Details

1) Data Set: Cityscape [36]—A Large Scale Street Views Data Set: It contains 26 948 training images with the size $H \times W = 1024 \times 2048$, where it includes 3475 fine annotated images, 3475 coarse annotated images, and 19 998 extra training images for 19-class semantic segmentation, and the corresponding disparity maps obtained by SGM [38] for stereo matching.

KITTI [7]—A Real-World Data Set With Street Views From a Driving Car: It consists of 200 training and 200 testing images with $H \times W = 376 \times 1240$, the same category as Cityscape [36] for semantic segmentation and the sparse ground-truth disparities obtained by LiDAR for stereo matching.

US3D [37], [39]—An RS Data Set With Incidental Satellite Images From Satellites: It includes 4292 training images and 50 testing images with the size $H \times W = 1024 \times 1024$. The classification labels obtained by the LSA specification for six-class semantic segmentation, and the sparse ground-truth disparities obtained by satellites for stereo matching.

Our results were submitted to the evaluation systems at the two websites for blind performance evaluation, as the systems implicitly provided test data with ground truth for public evaluation.

2) Training and Testing: We implement BGA-Net using TensorFlow and train the model with the ADAM optimizer ($\beta_1 = 0.9, \beta_2 = 0.999$) in an end-to-end manner. For the hyperparameters in the loss function, we first adopt $\lambda_1 = 1$, $\lambda_2 = 1$, $\lambda_3 = 1$, and $\lambda_4 = 1$ and then find the best parameters via our experiments. Before training, we normalize the input images with pixel intensities level ranging from 0 to 1.

[1] https://github.com/NPU-IAP/BGA-Net

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Due to different classification numbers and data types, we adopt one training way on the Cityscape and KITTI data sets and employ another way on the US3D data set.

1) For Cityscape [36] and KITTI [7], we set a batch size of 2, the disparity $D = 192$, and the number of category $C = 20$. We randomly crop the input images into 640×320 for Cityscape and 512×256 for KITTI. In the pretraining process, we trained our model from scratch using fine annotated images on Cityscape for 150 epochs. The learning rate for pretraining begins at 0.001 for the first 50 epochs and reduces to 0.0001 for the remaining epochs. In the fine-tuning process, we fine-tune the trained model on KITTI for 200 epochs using the learning rate 0.0001. Due to excessive memory usage, this training procedure is performed on two NVIDIA Titan V100 GPUs.
2) For US3D [37], [39], we set a batch size of 1, the disparity $D = 128$, and the number of category $C = 6$. We randomly crop the input images into 448×448, and the learning rate is initially set to 0.001 for 200 epochs and decreases to 0.0001 for the other 350 epochs. Note that this training procedure is performed on four NVIDIA 1080Ti GPUs.

For the testing process, we feed the full-size left–right images into the network and upload the final results to the related benchmark for evaluating our model.

3) Evaluation Metric: Given the two tasks, we follow the evaluation metrics as previous works [7], [36], [37]. In the segmentation task, we apply the mean intersection over union (mIoU) [36] to estimate the performance. The concrete calculating methods is represented as

$$\text{mIoU} = \frac{1}{C}\sum_{c}\frac{\text{TP}}{\text{TP}+\text{FP}+\text{FN}} \tag{8}$$

where TP means the matched pairs of segments, FP represents the unmatched predicted segments, FN indicates unmatched ground-truth segments, and C signifies the number of categories. In the stereo matching task, we employ the average endpoint error (EPE) and the fraction of erroneous pixels (D_1) [7] to assess the consistency. The metric is expressed as

$$\begin{aligned}\text{EPE} &= \frac{1}{N}\sum_{i\in T}|\hat{d}-d| \\ D_1 &= \frac{1}{N}\sum_{i\in T}[|\hat{d}-d|>t]\end{aligned} \tag{9}$$

where d and $\hat{d}$ denote the ground truth and estimated disparity, respectively, N and T represent the number and the set of labeled pixels in the image, respectively, $[\cdot]$ means the iverson bracket, and t indicates the threshold of error disparity. Note that t is set as 3 in general.

B. Ablations

To demonstrate the effectiveness of our method, we will decompose our approach step-by-step to reveal each component's effect via evaluating the performance of BGA-Net on the Cityscape and US3D data sets with different settings, including different architectures and different parameters of the loss function. First, we randomly divided the total fine annotated images into a training set with 3275 images and a validation set with 200 images on the Cityscape data set. Next, we adopt our NLCA-Net [34] and the DeepLab v3+ [26] as the baseline and then gradually transform NLCA-Net [34] into BGA-Net with the default hyperparameters $\lambda_1 = 1$, $\lambda_2 = 1$, $\lambda_3 = 1$, and $\lambda_4 = 1$. The effect of each component and the qualitative results of validation data are represented in Tables I and II and Fig. 7. Finally, we use the best structure to find the best hyperparameters on the Cityscape and US3D data set, as shown in Table III.

TABLE I

EVALUATION OF BGA-NET WITH DIFFERENT SETTINGS ON THE CITYSCAPE DATA SET. WE EMPLOY THE RECOGNIZED EVALUATION INDICATORS TO COMPARE THE EFFECTIVENESS OF FIVE MODULES ON THE CITYSCAPE DATA SET, INCLUDING UBM, BGAM, SSM, FMM, AND BFM

Network Setting					mIoU	EPE	D_1 (%)
UBM	BGAM	SSM	FFM	BFM			
		DeepLab	NLCA		0.776	0.546	3.475
√		DeepLab	√		0.776	0.527	3.391
√		√	√		0.764	0.577	3.673
√		√	√	√	0.623	1.432	9.474
√	√	√	√		0.783	0.448	2.126
√	√	√	√	√	**0.786**	**0.432**	**2.087**

TABLE II

COMPARED ASPP AND SPP OF OUR MODULE ON THE CITYSCAPE DATA SET

UBM		SSM		mIoU	D_1(%)
ASPP	SPP	ASPP	SPP		
-	-	-	-	0.767	2.91
√	-	√	-	0.783	2.10
√	-	-	√	**0.786**	**2.09**
-	√	√	-	0.784	2.27
-	√	-	√	0.779	2.54

As shown in Table I, the simple multitask framework performs slightly worse performance than the baseline, which consists of the UBM, SSM, and FMM. Due to without sharing the task-specific features, it causes the performance degradation somewhat either of the two tasks. More seriously, when we add the BFM into the framework, it may lead to unconvergent. It is caused by the impact of different loss functions and inadequate feature sharing. However, when we put the BGAM into the model, the network could be trained to convergent again and get the best performance than the previous results. It is proved that the task-specific feature sharing is a crucial factor in the MTL. Furthermore, the qualitative results also prove that our method could generate a more accurate segmentation map and precise disparity map for two tasks, as shown in Fig. 7.

As shown in Table II, we can get the conclusions as follows: 1) ASPP or SPP has a significant impact on semantic segmentation and stereo matching tasks, which has been demonstrated in previous work; 2) exchanging ASPP and SPP in UBM or SSM have a little effect on semantic segmentation; however, ASPP in UBM is more helpful to stereo matching, as enlarging

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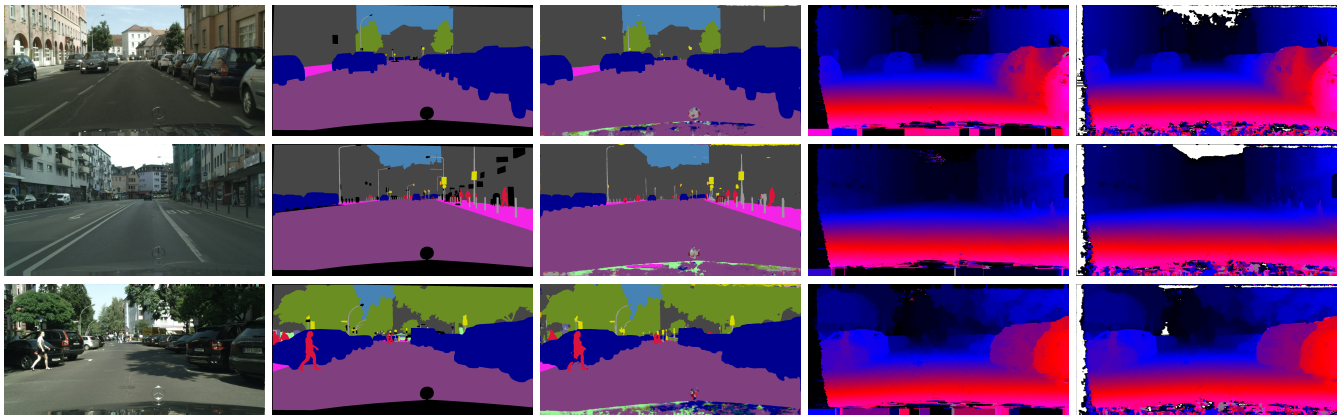

Fig. 7. Cityscape validation data qualitative results. Our approach can predict the segmentation map and the disparity map simultaneously and can be trained in an end-to-end manner. (a) Left image. (b) CLS GT. (c) CLS. (d) DSP GT. (e) DSP.

TABLE III

INFLUENCE OF WEIGHT VALUES FOR λ_1–λ_4 ON PERFORMANCE. WE EMPIRICALLY FOUND THAT 1.5/1.5/1/1 YIELDED THE BEST PERFORMANCE ON THE CITYSCAPE DATA SET AND 1.2/1.2/1/1 YIELDED THE BEST RESULT ON THE US3D DATA SET

Network setting				Cityscape		US3D	
λ_1	λ_2	λ_3	λ_4	mIoU	EPE	mIoU	EPE
1	1	1	1	0.786	0.432	0.764	1.431
0.5	1	0.5	1	0.774	0.421	0.761	1.448
1	1.5	1	1.5	0.775	0.422	0.774	1.333
1.5	1	1.5	1	0.789	0.453	0.777	1.306
2	2	1	1	0.792	0.483	0.779	1.276
1	1	1.2	1.2	0.783	0.413	0.776	1.252
1	1	0.8	0.8	0.784	0.439	0.768	1.269
0.8	0.8	1	1	0.754	0.419	0.776	1.235
1.2	1.2	1	1	0.791	0.433	**0.782**	**1.211**
1.5	1.5	1	1	**0.793**	**0.410**	0.780	1.224

TABLE IV

RESULTS ON THE KITTI 2015 STEREO BENCHMARK (∗ MEANS THE BGA-NET WITHOUT FINE-TUNING)

Method	*All pixels* (%)			*Non-Occluded pixels* (%)		
	D1-bg	D1-fg	D1-all	D1-bg	D1-fg	D1-all
L-ResMatch [40]	2.72 %	6.95 %	3.42 %	2.35 %	5.76 %	2.91 %
MC-CNN-acrt [41]	2.89 %	8.88 %	3.89 %	2.48 %	7.64 %	3.33 %
GC-NET [15]	2.21 %	6.16 %	2.87 %	2.02 %	5.58 %	2.61 %
PSMNet [16]	1.86 %	4.62 %	2.32 %	**1.71 %**	4.31 %	2.14 %
SsSMnet [42]	2.70 %	6.92 %	3.40 %	2.46 %	6.13 %	3.06 %
SGM-Net [43]	2.66 %	8.64 %	3.66 %	2.23 %	7.44 %	3.09 %
SPS-St [44]	3.84 %	12.67 %	5.31 %	3.50 %	11.61 %	4.84 %
Displets v2 [45]	3.00 %	5.56 %	3.43 %	3.43 %	4.46 %	3.09 %
PBCP [46]	2.58 %	8.78 %	3.61 %	2.27 %	7.71 %	3.17 %
BGA-Net*	4.56 %	13.43 %	8.44 %	4.29 %	12.76 %	7.63 %
BGA-Net	**1.85 %**	**4.09 %**	**2.22 %**	**1.71 %**	**3.93 %**	**2.08 %**

TABLE V

RESULTS ON KITTI SEMANTIC BENCHMARK (∗ MEANS THE BGA-NET WITHOUT FINE-TUNING)

Model	IoU class	iIoU class	IoU category	iIoU category
SDNet [48]	51.14 %	17.74 %	79.62 %	50.45 %
APMoE-seg [49]	47.96 %	17.86 %	78.11 %	49.17 %
BGA-Net*	53.14 %	21.19 %	78.23 %	51.15 %
BGA-Net	**59.24 %**	**28.12 %**	**81.51 %**	**59.99 %**

the receptive field is more effective in stereo matching; and 3) for semantic segmentation, ASPP either in UBM or SSM is just a little worse than SPP in SSM. Thus, we consider that multiscale feature maps can further improve performance when the receptive field is enough larger. To summarize, we decide to use ASPP in UBM and use SPP in SSM.

As shown in Table III, we continue to adjust the proportion of different loss functions based on experience for yielding the best results. Although we cannot test all the hyperparameters, it still could demonstrate that the different value of hyperparameters has dramatic effects on the performance. Through lots of tests and data analyses, we empirically find that 1.5/1.5/1/1 yielded the best performance on the Cityscape data set, and 1.2/1.2/1/1 yielded the best performance on the US3D data set. It also proves that due to the different classification numbers and the various data distribution on two data sets, we should use different hyperparameters to produce the best result.

C. KITTI

To evaluate our model's performance in traditional computer vision tasks, we compare our approach with other methods on the classical data set KITTI. First, we employ the fine-tuning model to produce the result of the test set. Then, we upload our result to KITTI server[2] for assessing the performance. The semantic segmentation and stereo matching results are represented as Tables IV and V, respectively. Moreover, we present the representative result of our model, as shown in Fig. 8.

As shown in Tables IV and V, it qualitatively demonstrates the performance of the proposed method compared with other state-of-the-art methods for the two tasks. The proposed approach generates more accurate segmentation and disparity

[2]http://www.cvlibs.net/datasets/kitti

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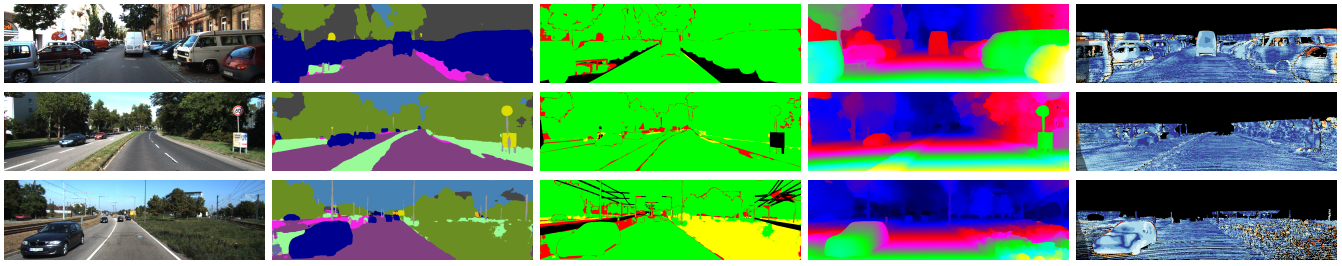

Fig. 8. Qualitative examples of BGA-Net on KITTI stereo and semantic test sets. By incorporating semantic information, our method is able to predict precise disparity maps. (a) Left image. (b) CLS. (c) CLS error map. (d) DSP. (e) DSP error map.

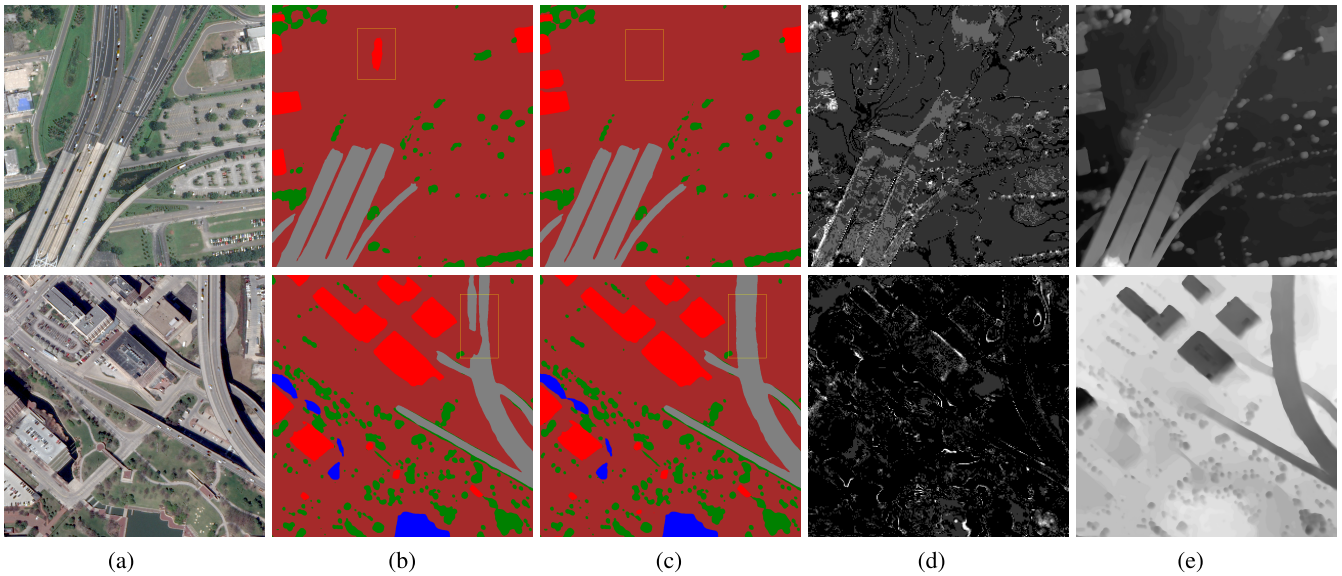

Fig. 9. Qualitative instances of the BGA-Net on US3D data set. With the fusion of disparity and segmentation maps, it avoids some absurd results in the segmentation task, e.g., confusion of irrelevant things and incomplete segmentation. Note that because the difference is minimal between DSP and fused DSP, we show the residual learning of our fusion module in (d). (a) Left image. (b) CLS. (c) Fused CLS. (d) Res-learning of DSP. (e) Fused DSP.

TABLE VI

PERFORMANCE COMPARISON ON THE US3D BENCHMARK. WE COMPUTED THE EPE, PERCENTAGE OF THREE-PIXEL ERROR, MIOU, AND MIOU-3 ON THE US3D TEST SET. HERE, $*$ MEANS USING TESTING DATA AUGMENTATION FOR SEMANTIC SEGMENTATION (ROTATION OR MIRROR IMAGE IN EIGHT DIRECTIONS AND THEN AVERAGE THEM) AND † DENOTES APPLYING POSTPROCESS LIKE [2] AND [3]

Model	mIoU	D_1 (%)	**EPE**	mIoU-3
ICNet [50] + SGM [51]	0.700	43.00	10.34	-
DeepLab v3 [26] + SGM [51]	0.750	43.00	10.34	-
ICNet [50] + iResNet-i2 [21]	0.700	33.00	3.05	-
DeepLab v3 [26] + iResNet-i2 [21]	0.750	33.00	3.05	-
ICNet [50] + DenseMapNet [52]	0.700	35.00	3.51	-
DeepLab v3 [26] + DenseMapNet [52]	0.750	35.00	3.51	-
SDBF-Net [33]	0.767	8.02	1.31	0.759
CU-PSM [2]	-	-	-	0.772
MRFCNet [3]	0.790	9.06	1.39	0.778
BGA-Net	0.782	7.26	1.21	0.772
BGA-Net*	0.790	7.26	1.21	0.781
BGA-Net†	**0.791**	**7.21**	**1.20**	**0.782**

maps than other methods. Moreover, our method without fine-tuning also achieves a good performance, which exhibits excellent generalization ability. Meanwhile, other state-of-the-art methods often are trained in the larger data set, such as

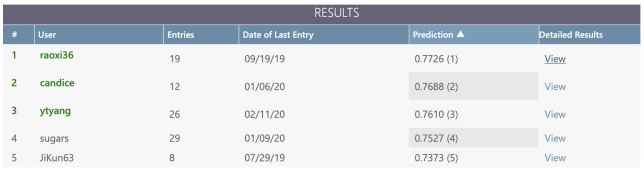

Fig. 10. Screenshot of our comprehensive score in the codalab server (id: raoxi36). The comprehensive score mIoU-3 contains two parts: mIoU and D_1.

SceneFlow [47]. However, we only use a relatively small data set (training data: 3475 versus 35 454 and inaccurate ground truth obtained by SGM [38]). Our result will be better if we have a more larger data set of 3-D semantic detection. As shown in Fig. 8, our approach yields clear disparity maps and outstanding segmentation maps, particularly in the nonoccluded region for the stereo matching task. Overall, BGA-Net provides a superior improvement over other state-of-the-art methods.

D. US3D

To evaluate our model's performance on US3D, we use the training method to train the model for calculating the

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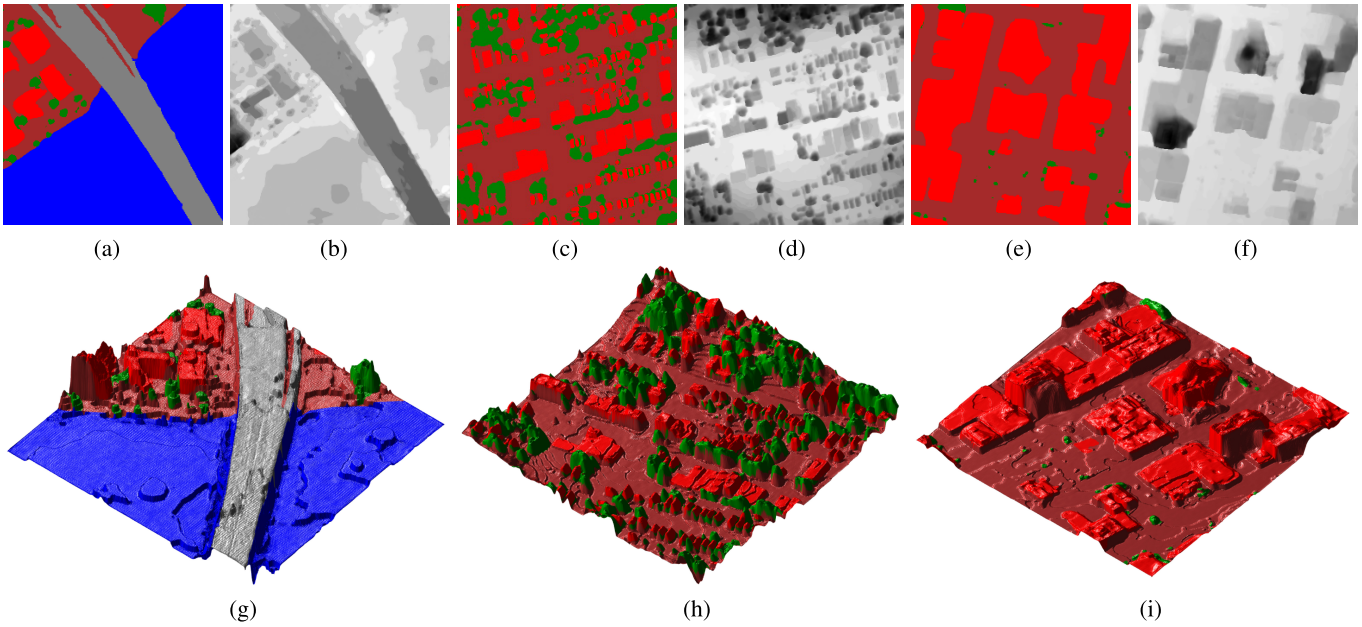

Fig. 11. Results of 3-D reconstruction. Given the segmentation and disparity map [e.g., (a) and (b), (c) and (d), or (e) and (f)], we only use the simple surface reconstruction method of MATLAB to obtain the 3-D model in (g)–(i). It shows that our model can be used in the large 3-D scene reconstruction. (a), (c), and (e) CLS. (b), (d), and (f) DSP. (g)–(i) 3-D reconstruction

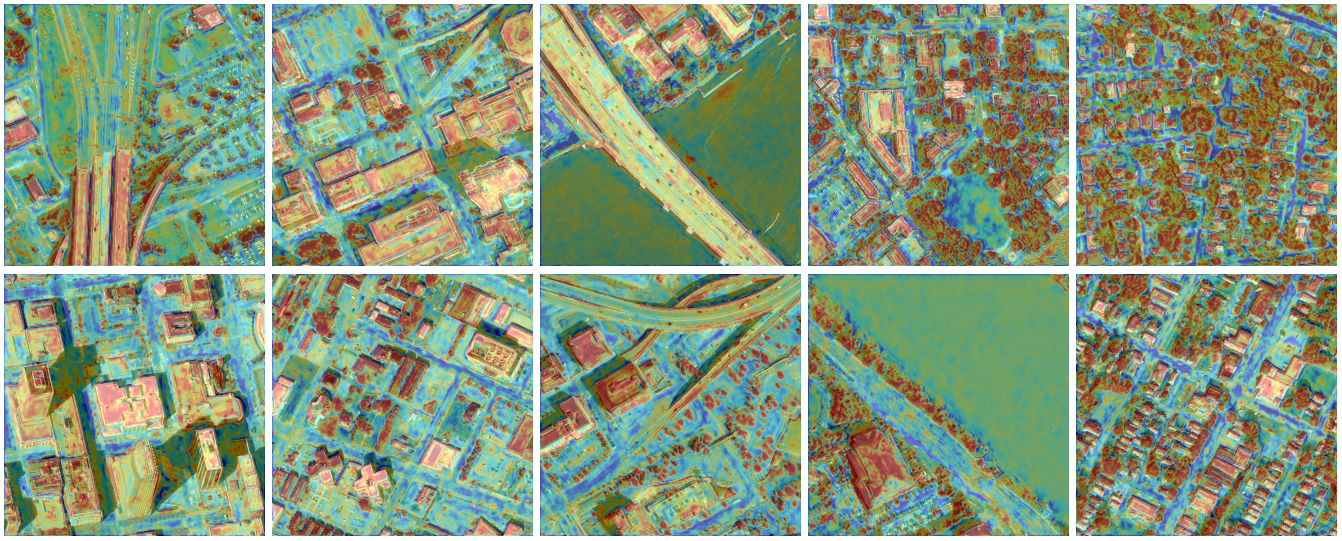

Fig. 12. Heatmaps of segmentation attention map for different images on the US3D data set (the red regions are assigned more weights, whereas the blue areas fewer weights). By introducing the global information from GC-block, our attention mechanism exhibits a strong ability to observe the target object in the image.

segmentation and disparity maps, as shown in Section IV-A. Then, we upload the results to the CodaLab evaluation server. Because US3D is an emerging data set and does not attach the related papers in the CodaLab, we adopt the baseline results provided in [37] as a reference to assess our performance in Table VI and present our high-quality results in Fig. 9.

As presented in Table VI and Fig. 9, our approach achieves a top-class performance in the CodaLab evaluation server[3] (id: raoxi36, the screenshot of our comprehensive score mIoU-3 as shown in Fig. 10). Moreover, the mIoU, EPE, and D_1 scores of our method have been superior to the baseline or other methods since June 29, 2019, ranking the Top First by the overall score. The estimated results demonstrate that BGA-Net significantly improves the completeness and precision for the 3-D semantic detection. Compared with CU-PSM [2] and MRFCNet [3], our method can directly output outstanding results without any postprocess. If we apply the same testing strategy and postprocess as they do, we can also obtain a more significant improvement in semantic segmentation because our results of disparity are better than them.

Our method takes full advantage of visual attention technique to largely promote various indicators of results compared with other state-of-the-art methods. Applying attention

[3]https://competitions.codalab.org/competitions/20212#results

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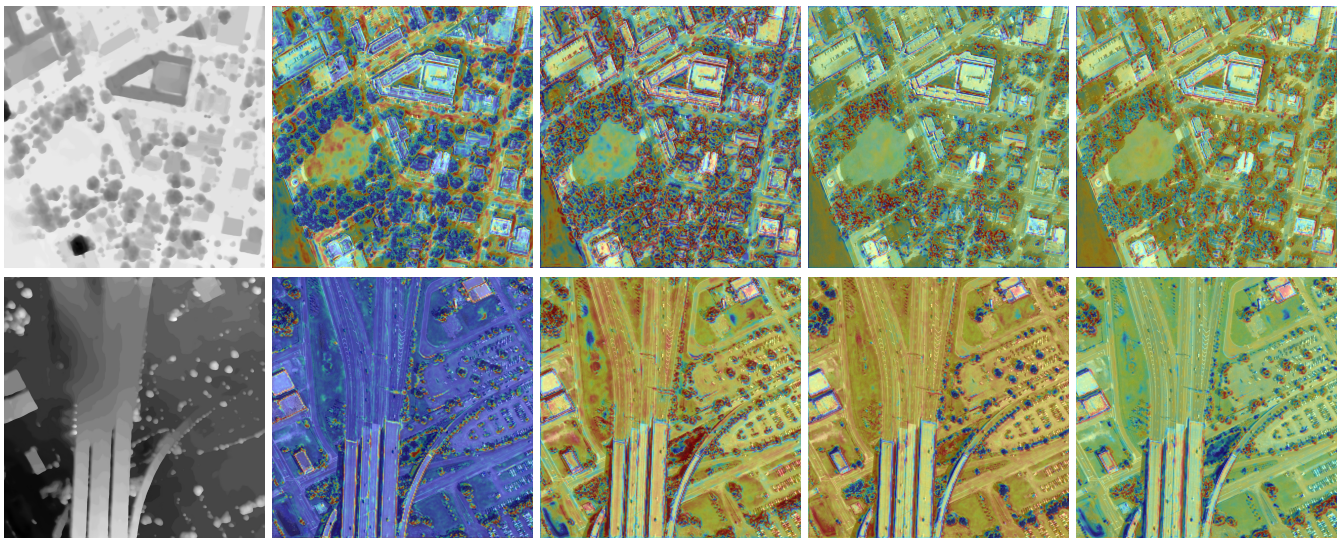

Fig. 13. Heat-maps of disparity attention map for different images on the US3D data set (the red regions are assigned more weights, whereas the blue areas fewer weights). By our attention mechanism, the network could predict the probable target object for the different disparities. (From left) Disparity map and attention maps with four different disparity values. (a) Left image. (b) $d = 0$. (c) $d = 4$. (d) $d = 8$. (e) $d = 16$.

strategy as a connection between two tasks, it enables to perform the two tasks in a model and drives the model to have the state-of-the-art results for the two tasks. Meanwhile, as adding the labels of segmentation task, it can enhance the scene understanding ability of the model and make our model handle well against the seasonal appearance differences and get more accurate results in the matching task. Moreover, we present our results of 3-D reconstruction in Fig. 11, which indicates that our model can be easily applied in the large 3-D scene reconstruction of RS images.

E. Bidirectional Guided Attention Module Analysis

To verify the reliability of the proposed approach, we apply our best model in US3D to produce the semantic and disparity attention maps. For a clear presentation, we enlarge the attention map to the original size of input images via the bilinear interpolation and then generate the heatmaps by combining the input images. The semantic and disparity attention maps are shown in Figs. 12 and 13, respectively.

As shown in Fig. 12, the semantic attention maps could allocate more weight to the target object. It illustrates that the proposed structure could make the SSM focus on the foreground object, which brings more contextual information to the whole network for improving the scene understanding ability. As shown in Fig. 13, the group of disparity attention maps could assign different weights for each disparity. It indicates that the proposed architecture could pay attention to the target object and assign it to the possible disparity. In short, BGAM extracts the generalizable representation for two tasks and improves the feature sharing ability, which brings a remarkable improvement and advances for MTL.

F. Season Challenge Analysis in Stereo Matching

To analyze our huge advantages in stereo matching, we do a contrast experience to prove the necessity of a UBM between two tasks. We consider that scene understanding is crucial to deal with seasonal appearance differences, making networks extract more general features. Thus, we employ a UBM to extract features and share these features into three branches. As we know, the neural network is a black box, and we cannot understand the specific meaning of features. Therefore, we only prove it indirectly. First, we pass the left unified feature maps X_l to our SSM and then can get a CLS of the left image. Next, we feed the right unified feature maps X_r to SSM and obtain a CLS of the right image. The results are presented in Fig. 14.

As shown in Fig. 14, our CLS of the right image is also pretty good. It may indirectly demonstrate that our unified feature maps contain wealthy semantic information. We consider that rich semantic information can help the network to understand the scene and make networks cope with seasonal appearance differences. Moreover, compared with the single stereo model in Tables I and VI, our model achieves a significant benefit in stereo matching, which proves its effectiveness. Obviously, shared feature extraction not only reduces the number of parameters but also promotes performance improvement and end-to-end prediction. As all right images have no ground truth, as a result, quantitative comparison results cannot be given. However, intuitively, we think that the result of the right image is also excellent.

V. Conclusion

In this work, we have presented a novel MTL approach for 3-D semantic detection, named the BGA-Net. The critical point of the proposed method is to employ the GC-block to build the bidirectional (bottom–up and up–bottom) visual attention mechanism and then apply the generated attention maps to guide the learning process for sharing task-specific features. Moreover, compared with other MTL methods, our network can be trained in an end-to-end manner and

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TABLE VII
SUMMARY OF OUR BGA-NET

No.	Layer Description (k, s, f)	Output dimension
	Input images	$H \times W$
Unified Backbone Module (UBM)		
conv0_1	$3 \times 3, 2, 32$	$1/2H \times 1/2W \times F$
conv0_2	$\begin{bmatrix} 3 \times 3, 1, 32 \end{bmatrix} \times 3$	$1/2H \times 1/2W \times F$
conv0_3	$\begin{bmatrix} 3 \times 3, 1, 64 \\ 3 \times 3, 1, 64 \end{bmatrix} \times 3$	$1/2H \times 1/2W \times 2F$
conv0_4	$3 \times 3, 2, 128$ $\begin{bmatrix} 1 \times 1, 1, 64 \\ 3 \times 3, 1, 64 \\ 3 \times 3, 1, 128 \end{bmatrix} \times 16$	$1/4H \times 1/4W \times 4F$
conv0_5	$3 \times 3, 1, 192$ $\begin{bmatrix} 3 \times 3, 192, dila = 2 \\ 3 \times 3, 192, dila = 2 \end{bmatrix}$	$1/4H \times 1/4W \times 6F$
conv0_6	$3 \times 3, 1, 256$ $\begin{bmatrix} 3 \times 3, 256, dila = 4 \\ 3 \times 3, 256, dila = 4 \end{bmatrix}$	$1/4H \times 1/4W \times 8F$
ASPP	$\begin{bmatrix} 1 \times 1, 1, 64 \\ 3 \times 3, 64, dila = 6 \\ 3 \times 3, 64, dila = 12 \\ 3 \times 3, 64, dila = 18 \end{bmatrix}$	$1/4H \times 1/4W \times 8F$
	concat conv0_4, conv0_5, conv0_6, and ASPP	$1/4H \times 1/4W \times 26F$
conv0_7	$3 \times 3, 1, 768$ $3 \times 3, 1, 512$ (without ReLU and BN)	$1/4H \times 1/4W \times 16F$
Bidirectional Attention module (BAM)		
trans0	$1 \times 1, 1, 256$ $3 \times 3, 1, 128$ $\begin{bmatrix} 3 \times 3, 1, 128 \\ 3 \times 3, 1, 128 \end{bmatrix}$	$1/4H \times 1/4W \times 4F$
conv1_0	GC-block $\begin{bmatrix} 3 \times 3, 1, 128 \\ 3 \times 3, 1, 128 \end{bmatrix}$	$1/4H \times 1/4W \times 4F$
conv1_1 - conv1_3	repeat conv1_0	$1/4H \times 1/4W \times 4F$
cls_attention	$1 \times 1, 1, 64$ $1 \times 1, 1, 1$ (without ReLU and BN) sigmoid	$1/4H \times 1/4W \times 1$
dsp_attention	$1 \times 1, 1, 64$ $1 \times 1, 1, 1/4D$ (without ReLU and BN) sigmoid	$1/4H \times 1/4W \times 1/4D$
Semantic Segmentation module (SSM))		
trans1	same structure as trans0	$1/4H \times 1/4W \times 4F$
conv2_0	$\begin{bmatrix} 1 \times 1, 1, 64 \\ 3 \times 3, 1, 64 \\ 3 \times 3, 1, 128 \end{bmatrix} \times 16$	$1/4H \times 1/4W \times 4F$
SPP	$\begin{bmatrix} ave_pooling : 64, 1 \times 1, 1, 32 \\ ave_pooling : 32, 1 \times 1, 1, 32 \\ ave_pooling : 16, 1 \times 1, 1, 32 \\ ave_pooling : 8, 1 \times 1, 1, 32 \end{bmatrix}$ bilinear interpolation	$1/4H \times 1/4W \times 4F$
	concat conv2_0 and SPP	$1/4H \times 1/4W \times 8F$
conv2_1	$3 \times 3, 1, 128$ $3 \times 3, 1, 64$	$1/4H \times 1/4W \times 2F$
	cls_attention $\cdot$ conv2_1 + conv2_1	$1/4H \times 1/4W \times 2F$
deconv2_0	deconv: $3 \times 3, 2, 32$ deconv: $3 \times 3, 2, C$ (without ReLU and BN)	$H \times W \times C$
conv2_2	$3 \times 3, 1, C$ (without ReLU and BN)	$H \times W \times C$
Feature Matching module (FMM)		
trans2	$1 \times 1, 1, 128$ $3 \times 3, 1, 64$	$1/4H \times 1/4W \times 2F$
	construct cost volume	$1/4D \times 1/4H \times 1/4W \times 2F$
conv3_0	$3 \times 3 \times 3, 1, 64$ $3 \times 3 \times 3, 1, 32$ $\begin{bmatrix} 3 \times 3 \times 3, 1, 32 \\ 3 \times 3, 1, 32 \end{bmatrix}$	$1/4D \times 1/4H \times 1/4W \times F$
conv3_1 - conv3_4	$3 \times 3 \times 3, 2, (i+1) \times 32$ $\begin{bmatrix} 3 \times 3 \times 3, 1, (i+1) \times 32 \\ 3 \times 3 \times 3, 1, (i+1) \times 32 \end{bmatrix}, i = 1, 2, 3, 4$	$\frac{1}{2^{(i+2)}} D \times \frac{1}{2^{(i+2)}} H \times \frac{1}{2^{(i+2)}} W \times (i+1)F$
conv3_5	NL-block $\times$ 4	$1/64D \times 1/64H \times 1/64W \times 5F$
deconv3_3 - deconv3_0	deconv: $3 \times 3 \times 3, 2, (i+1) \times 32$ $\begin{bmatrix} 3 \times 3 \times 3, 1, (i+1) \times 32 \\ 3 \times 3 \times 3, 1, (i+1) \times 32 \end{bmatrix}, i = 3, 2, 1, 0$ add conv3_i	$\frac{1}{2^{(i+2)}} D \times \frac{1}{2^{(i+2)}} H \times \frac{1}{2^{(i+2)}} W \times (i+1)F$
	dsp_attention $\cdot$ deconv3_0 + deconv3_0	$1/4H \times 1/4W \times 2F$
deconv3_4	deconv: $3 \times 3 \times 3, 2, 16$ deconv: $3 \times 3 \times 3, 2, 1$ (without ReLU and BN)	$D \times H \times W \times 1$
conv3_6	$3 \times 3 \times 3, 1, 1$ (without ReLU and BN)	$D \times H \times W \times 1$
	Soft argmin	$H \times W$
Fusion module (FM)		
	concat left image, disparity map, and score map for the fusion of segmentation map. concat left image, disparity map, and segmentation map for the fusion of disparity map.	$H \times W \times (C+4)$ or $H \times W \times 5$
conv4_0	$3 \times 3, 1, 32$	$H \times W \times F$
conv4_1- conv4_4	$3 \times 3, 2, (i+1) \times 32, i = 1, 2, 3, 4$	$1/2^i H \times 1/2^i W \times (i+1)F$
deconv4_3-conv4_0	deconv: $3 \times 3, 2, (i+1) \times 32$ add conv4_i $, i = 3, 2, 1, 0$	$1/2^i H \times 1/2^i W \times (i+1)F$
conv4_5	$3 \times 3, 1, C$ (without ReLU and BN) for segmentation map, add initial score map $3 \times 3, 2, 1$ (without ReLU and BN) for disparity map, add initial disparity map	$H \times W \times C$ or $H \times W$

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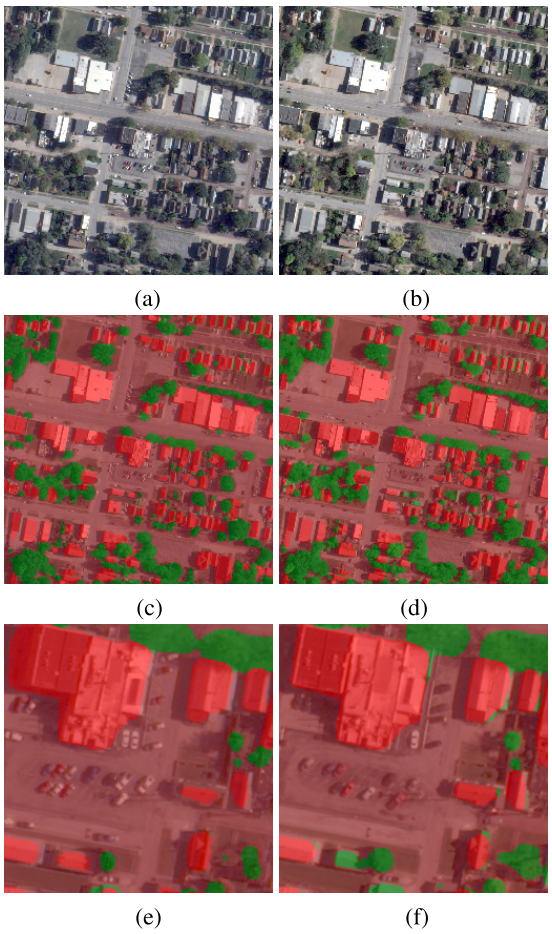

Fig. 14. Bidirectional: the CLS results of (a) left and (b) right images. We feed left and right unified feature maps $X_{l,r}$ to SSM, and then obtain CLS results. Through comparing CLS results of the left and right images, it is clear that our unified feature maps contain wealthy semantic information. Combined with results in the stereo matching task, we consider that scene understanding is helpful to handle seasonal appearance differences in RS images. (c) Left CLS. (d) Right CLS. (e) Part of left CLS. (f) Part of right CLS.

generate the segmentation and disparity maps simultaneously. Experiments on the Cityscape, KITTI, and US3D data sets show that our method can generate the state-of-the-art results for the two tasks and has extraordinary robustness to seasonal appearance differences on incidental satellite images, which significantly outperforms other competitive methods. Furthermore, compared with the single-task models, our model's results can be directly used to reconstruct the 3-D scene of RS images and get a nice visual effect.

APPENDIX
DETAILED NETWORK STRUCTURE

The core architecture of our BGA-Net contains five modules: 1) UBM; 2) bidirectional attention module; 3) SSM; 4) FMM; and 5) BFM. We provide the detailed structure of our approach, as presented in Table VII. Each 2-D or 3-D convolutional layer contains three steps: convolution, batch normalization (BN), and ReLU nonlinearity (unless otherwise specified).

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